

Investigating Windows Event Logs

Security incident analysis with Event Viewer, Log Parser and PowerShell

Exercises 1–5 | Security log | Event IDs 4625 and 4624

05

**From raw events
to a defensible
incident assessment**

ACCESS • FILTER • PARSE • CORRELATE

The workflow progresses from access to correlation

- | | | |
|----|-----------------------------|--|
| 01 | Access | Open Event Viewer and navigate to the Windows Security log. |
| 02 | Filter & export | Isolate failed logons (4625) and preserve them as an EVTX file. |
| 03 | Parse & automate | Extract records with Log Parser and confirm them with PowerShell. |
| 04 | Correlate | Compare failed logons with successful logons (4624) and report limits. |

The analysis uses three complementary Windows tools

```
PS C:\WINDOWS\system32> logparser -h
logparser : The term 'logparser' is not recognized as the name of a cmdlet, function, script file, or operable
program. Check the spelling of the name, or if a path was included, verify that the path is correct and try a
At line:1 char:1
+ logparser -h
+ ~~~~~
+ CategoryInfo          : ObjectNotFound: (logparser:String) [], CommandNotFoundException
+ FullyQualifiedErrorId : CommandNotFoundException

PS C:\WINDOWS\system32> $env:Path += ";C:\Program Files (x86)\Log Parser 2.2"
PS C:\WINDOWS\system32> logparser -h

Microsoft (R) Log Parser Version 2.2.10
Copyright (C) 2004 Microsoft Corporation. All rights reserved.

Usage:   LogParser [-i:<input_format>] [-o:<output_format>] <SQL query> |
         file:<query_filename>[?param1=value1+...]
         [<input_format_options>] [<output_format_options>]
         [-q[:ON|OFF]] [-e:<max_errors>] [-iw[:ON|OFF]]
         [-stats[:ON|OFF]] [-saveDefaults] [-queryInfo]

LogParser -c -i:<input_format> -o:<output_format> <from_entity>
```

Event Viewer

Browse, filter and export Windows Security records.

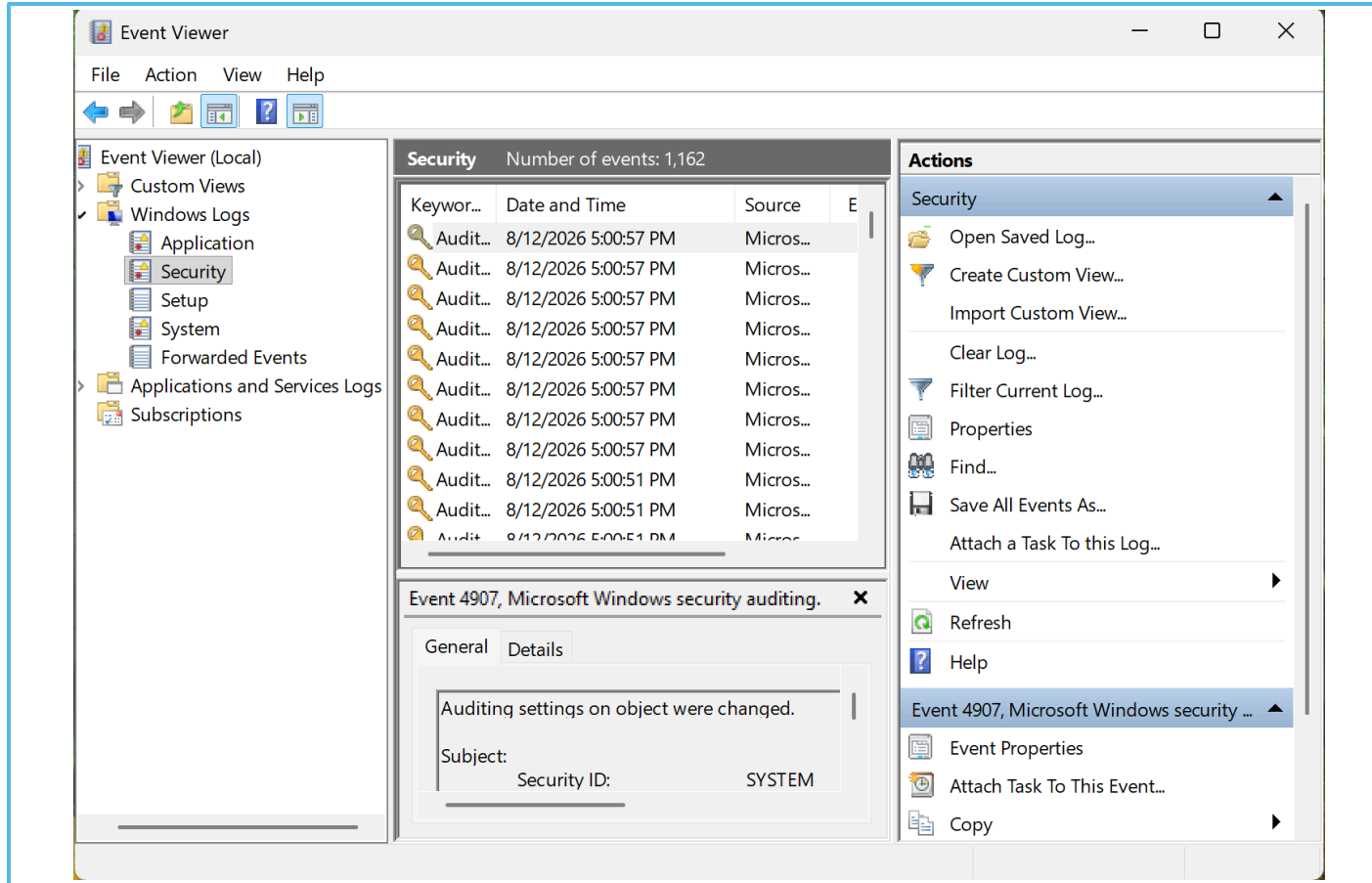
Log Parser 2.2

Convert EVTX evidence into structured CSV output.

PowerShell

Query Event IDs directly and validate the result.

Exercise 1 confirms access to the Security event log



Observed evidence

The Security log opened successfully and initially contained 1,162 records.

Why it matters

The Security channel records authentication and audit activity used throughout the remaining exercises.

Result: Event Viewer was ready for targeted review of logon activity.

Exercise 2 isolates five failed logon attempts

Security Number of events: 1,162					
Filtered: Log: Security; Source: ; Event ID: 4625. Number of events: 5					
Keywor...	Date and Time		Source	Event ID	Task Ca...
 Audit...	8/12/2026 5:26:26 PM		Micros...	4625	Logon
 Audit...	8/12/2026 5:25:53 PM		Micros...	4625	Logon
 Audit...	8/12/2026 5:25:40 PM		Micros...	4625	Logon
 Audit...	8/12/2026 5:25:37 PM		Micros...	4625	Logon
 Audit...	8/12/2026 5:25:35 PM		Micros...	4625	Logon

Finding

Five Event ID 4625 records were generated between 17:25:35 and 17:26:26 on 12 August 2026.

Preservation

The filtered records were exported as FailedLogins.evtx for repeatable analysis.

Exercise 3 converts the EVTX evidence into CSV

```
Command Prompt
Microsoft Windows [Version 10.0.26200.8875]
(c) Microsoft Corporation. All rights reserved.

C:\Users\rileyworth3>cd "C:\Program Files (x86)\Log Parser 2.2"

C:\Program Files (x86)\Log Parser 2.2>LogParser.exe "SELECT TimeGenerated, EventID, EventTypeName, Message INTO C:\Users\rileyworth3\Desktop\FailedLogins.csv FROM C:\Users\rileyworth3\Desktop\FailedLogins.evtx" -i:EVT -o:CSV

Statistics:
-----
Elements processed: 5
Elements output:    5
Execution time:     0.04 seconds

C:\Program Files (x86)\Log Parser 2.2>
```

```
TimeGenerated,EventID,EventTypeName,Message
2026-08-12 17:25:35,4625,Failure Audit event,The description for Event ID 4625 in Source "Microsoft-Windows-Security-Auditing"
cannot be found. The local computer may not have the necessary registry information or message DLL files to display messages from a
remote computer
2026-08-12 17:25:37,4625,Failure Audit event,The description for Event ID 4625 in Source "Microsoft-Windows-Security-Auditing"
cannot be found. The local computer may not have the necessary registry information or message DLL files to display messages from a
remote computer
2026-08-12 17:25:40,4625,Failure Audit event,The description for Event ID 4625 in Source "Microsoft-Windows-Security-Auditing"
cannot be found. The local computer may not have the necessary registry information or message DLL files to display messages from a
remote computer
2026-08-12 17:25:53,4625,Failure Audit event,The description for Event ID 4625 in Source "Microsoft-Windows-Security-Auditing"
cannot be found. The local computer may not have the necessary registry information or message DLL files to display messages from a
remote computer
2026-08-12 17:26:26,4625,Failure Audit event,The description for Event ID 4625 in Source "Microsoft-Windows-Security-Auditing"
cannot be found. The local computer may not have the necessary registry information or message DLL files to display messages from a
remote computer
```

Validated result

5 processed
5 output
0.04 seconds

Limitation

Timestamps and Event IDs were retained; message text did not resolve.

Exercise 4 independently reproduces the five failures

```
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

Install the latest PowerShell for new features and improvements! https://aka.ms/PSWindows

PS C:\Users\rileyworth3> Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4625} | Format-Table TimeCreated, Id, Message -AutoSize
```

TimeCreated	Id	Message
-----	--	-----
8/12/2026 5:26:26 PM	4625	An account failed to log on....
8/12/2026 5:25:53 PM	4625	An account failed to log on....
8/12/2026 5:25:40 PM	4625	An account failed to log on....
8/12/2026 5:25:37 PM	4625	An account failed to log on....
8/12/2026 5:25:35 PM	4625	An account failed to log on....

Command result

Get-WinEvent returned the same five timestamps and identified each record as “An account failed to log on.”

Cross-tool validation

Event Viewer, Log Parser and PowerShell agree on the event count and time sequence.

Exercise 5 gathers successful logons for comparison

Event Viewer (Local)

> Custom Views

▼ Windows Logs

Application

Security

Setup

System

Forwarded Events

> Applications and Services Logs

Subscriptions

SecurityNumber of events: 1,465 (!) New events available

Filtered: Log: Security; Source: ; Event ID: 4624. Number of events: 303

Keywords	Date an...	Source	Event...	Task Category
Audit S...	8/12/20...	Microsoft Wi...	4624	Logon
Audit S...	8/12/20...	Microsoft Wi...	4624	Logon
Audit S...	8/12/20...	Microsoft Wi...	4624	Logon
Audit S...	8/12/20...	Microsoft Wi...	4624	Logon
Audit S...	8/12/20...	Microsoft Wi...	4624	Logon
Audit S...	8/12/20...	Microsoft Wi...	4624	Logon

Observed evidence

The 4624 filter contained 303 records. The visible sample includes routine system and service logons.

Important distinction

A high 4624 count is not automatically suspicious: Windows creates many service, startup and local authentication events.

The comparison does not demonstrate a matching successful login

```
C:\Users\rileyworth3\Desktop>findstr "4625" CorrelatedLogins.csv
2026-08-12 17:25:35,4625,S-1-5-18|RILEYWORTH5FC$|WORKGROUP|0x3e7|S-1-0-0|forensic_test|RILEYWORTH5FC|0xc000006d|%%2313
|0xc000006a|2|User32|Negotiate|RILEYWORTH5FC|-|-|0|0x598|C:\Windows\System32\svchost.exe|127.0.0.1|0
2026-08-12 17:25:37,4625,S-1-5-18|RILEYWORTH5FC$|WORKGROUP|0x3e7|S-1-0-0|forensic_test|RILEYWORTH5FC|0xc000006d|%%2313
|0xc000006a|2|User32|Negotiate|RILEYWORTH5FC|-|-|0|0x598|C:\Windows\System32\svchost.exe|127.0.0.1|0
2026-08-12 17:25:40,4625,S-1-5-18|RILEYWORTH5FC$|WORKGROUP|0x3e7|S-1-0-0|forensic_test|RILEYWORTH5FC|0xc000006d|%%2313
|0xc000006a|2|User32|Negotiate|RILEYWORTH5FC|-|-|0|0x598|C:\Windows\System32\svchost.exe|127.0.0.1|0
2026-08-12 17:25:53,4625,S-1-5-18|RILEYWORTH5FC$|WORKGROUP|0x3e7|S-1-0-0|forensic_test|RILEYWORTH5FC|0xc000006d|%%2313
|0xc000006a|2|User32|Negotiate|RILEYWORTH5FC|-|-|0|0x598|C:\Windows\System32\svchost.exe|127.0.0.1|0
2026-08-12 17:26:26,4625,S-1-5-18|RILEYWORTH5FC$|WORKGROUP|0x3e7|S-1-0-0|forensic_test|RILEYWORTH5FC|0xc000006d|%%2313
|0xc000006a|2|User32|Negotiate|RILEYWORTH5FC|-|-|0|0x598|C:\Windows\System32\svchost.exe|127.0.0.1|0
```

FAILED — 4625

forensic_test
17:25:35–17:26:26
Source shown: 127.0.0.1

```
C:\Users\rileyworth3\Desktop>findstr "4624" CorrelatedLogins.csv
2026-08-12 21:04:35,4624,S-1-0-0|-|-|0x0|S-1-5-18|SYSTEM|NT AUTHORITY|0x3e7|0|-|-|0|00000000-0000-0000-0000-000000000000
0|-|-|0|0x4|-|-|0|-|-|0|0x1843|0x0|%%1842
2026-08-12 21:04:35,4624,S-1-5-18|RILEY-7T58EJFQC$|0x3e7|S-1-5-18|SYSTEM|NT AUTHORITY|0x3e7|5|Advapi|Negotiate|-|000
00000-0000-0000-0000-000000000000|-|-|0|0x370|C:\Windows\System32\services.exe|-|-|0|0x1833|-|-|0|0x1843|0x0|%%1842
2026-08-12 21:04:35,4624,S-1-5-18|RILEY-7T58EJFQC$|0x3e7|S-1-5-96-0-0|UMFD-0|Font Driver Host|0x529e2|2|Advapi|Negoti
ate|-|00000000-0000-0000-0000-000000000000|-|-|0|0x2dc|C:\Windows\System32\wininit.exe|-|-|0|0x1833|-|-|0|0x1842|0x0|%%
1843
2026-08-12 21:04:35,4624,S-1-5-18|RILEY-7T58EJFQC$|0x3e7|S-1-5-96-0-1|UMFD-1|Font Driver Host|0x529f3|2|Advapi|Negoti
ate|-|00000000-0000-0000-0000-000000000000|-|-|0|0x324|C:\Windows\System32\winlogon.exe|-|-|0|0x1833|-|-|0|0x1842|0x0|%%
1843
2026-08-12 21:04:35,4624,S-1-5-18|RILEY-7T58EJFQC$|0x3e7|S-1-5-20|NETWORK SERVICE|NT AUTHORITY|0x3e7|5|Advapi|Negotia
te|-|00000000-0000-0000-0000-000000000000|-|-|0|0x370|C:\Windows\System32\services.exe|-|-|0|0x1833|-|-|0|0x1843|0x0|%%
1842
2026-08-12 21:04:36,4624,S-1-5-18|RILEY-7T58EJFQC$|0x3e7|S-1-5-18|SYSTEM|NT AUTHORITY|0x3e7|5|Advapi|Negotiate|-|000
00000-0000-0000-0000-000000000000|-|-|0|0x370|C:\Windows\System32\services.exe|-|-|0|0x1833|-|-|0|0x1843|0x0|%%1842
2026-08-12 21:04:36,4624,S-1-5-18|RILEY-7T58EJFQC$|0x3e7|S-1-5-18|SYSTEM|NT AUTHORITY|0x3e7|5|Advapi|Negotiate|-|000
00000-0000-0000-0000-000000000000|-|-|0|0x370|C:\Windows\System32\services.exe|-|-|0|0x1833|-|-|0|0x1843|0x0|%%1842
2026-08-12 21:04:36,4624,S-1-5-18|RILEY-7T58EJFQC$|0x3e7|S-1-5-18|SYSTEM|NT AUTHORITY|0x3e7|5|Advapi|Negotiate|-|000
00000-0000-0000-0000-000000000000|-|-|0|0x370|C:\Windows\System32\services.exe|-|-|0|0x1833|-|-|0|0x1843|0x0|%%1842
2026-08-12 21:04:36,4624,S-1-5-18|RILEY-7T58EJFQC$|0x3e7|S-1-5-90-0-1|DWM-1|Window Manager|0x595f3|2|Advapi|Negotiate
|-|00000000-0000-0000-0000-000000000000|-|-|0|0x324|C:\Windows\System32\winlogon.exe|-|-|0|0x1833|-|-|0|0x1842|0x5960e
|%%1842
```

SUCCESSFUL — 4624

Visible sample: SYSTEM/service accounts
Around 21:04:35 onward
No matching forensic_test success shown

Defensible conclusion

The screenshots prove controlled failures, not a later success for the same account.

The evidence supports a controlled authentication test

Evidence	Observation	Interpretation	Confidence
Event Viewer	5 × 4625	Five failed logon attempts	High
Log Parser	5 processed / 5 output	EVTX successfully structured as CSV	High
PowerShell	Same five timestamps	Independent validation of failures	High
Event Viewer	303 × 4624	Many routine successful logons present	High
Correlation	No matching success shown	No confirmed account compromise	Moderate

Future analysis should correlate fields, not only event counts

1

Preserve originals

Retain the EVTX files and calculate hashes before further processing.

2

Use structured fields

Extract TargetUserName, IpAddress, LogonType, Status and SubStatus.

3

Apply a time window

Compare 4625 and 4624 records for the same account and source within minutes.

4

Separate test activity

Record the exact test account and test period so deliberate events are identifiable.

5

Broaden the timeline

Review 4634, 4648 and 4672 alongside process and network evidence.

Key lesson

Combining two CSV files creates one searchable dataset, but a genuine correlation must match account, source, logon type and time.

Five failed logons detected. No compromise confirmed.

Event Viewer isolated five 4625 events for the controlled test. Log Parser and PowerShell reproduced the same sequence. Although 303 successful 4624 events were available, the visible evidence did not link a later success to the forensic_test account.

Final assessment: expected lab-generated failures; continue field-level correlation before alleging an incident.